

ALEXIS

Trial Update Categorization System

Detection, classification, and outcome assessment for clinical trial changes

13 — update categories across 3 impact tiers

All — changes tagged with direction (up/down/from/to)

Phase linkage — detects drug program advancement across NCT IDs

Outcome assessment — completed trials scored as passed / failed / unknown

8-step pipeline — from raw pulse to enriched snapshot + changelog report

Source: ALEXIS Pulse snapshots diffed against Master DB (active universe)

Date: 2026-02-24

1. Update Category Taxonomy

Every pulse trial gets one or more of these tags. Multiple categories per trial are allowed — a trial can be `status_to_recruiting` AND `enrollment_change:enrollment_up` in the same week.

TIER 1 — HIGH IMPACT

Category	What Changed	Detection Method	Teams	Business Signal
<code>new_registration</code>	Trial not in master DB (new NCT ID)	Pulse NCT ID not found in master DB index	BD, Ops	Brand new opportunity. Sponsor actively placing contracts.
<code>phase_advance</code>	New NCT ID for same drug at later phase	Linkage engine: sponsor + drug + condition match (see §3)	BD, Ops, Sci	Sample volume escalation. Method validation increase. High-probability RFP.
<code>status_to_recruiting</code>	<code>overallStatus</code> → <code>RECRUITING</code>	Diff <code>overallStatus</code> : master ≠ pulse, pulse = <code>RECRUITING</code>	BD, Ops	Trial is live. Bioanalytical work imminent. Urgent if no contract exists.
<code>status_to_completed</code>	<code>overallStatus</code> → <code>COMPLETED</code>	Diff <code>overallStatus</code> : master ≠ pulse, pulse = <code>COMPLETED</code>	BD, Ops, Mkt	Revenue wind-down. If passed → expect phase advance. If failed → program likely dead.
<code>status_to_stopped</code>	Status → <code>TERMINATED</code> / <code>WITHDRAWN</code> / <code>SUSPENDED</code>	Diff <code>overallStatus</code> against known stop states	BD, Ops	Contract at risk. Analyst capacity unexpectedly freed.
<code>new_arm_added</code>	New intervention entry appeared	Diff intervention list: pulse has entries not in master	BD, Sci, Ops	New molecule may need new assay. Check if different modality.

TIER 2 — MEDIUM IMPACT

Category	What Changed	Detection Method	Teams	Business Signal
<code>enrollment_change</code>	Target enrollment increased or decreased	Diff <code>enrollmentInfo.count</code> : master ≠ pulse	Ops, BD	Up = more samples, longer timeline. Down = scope reduction, possible trouble.
<code>intervention_changed</code>	Drug name, dose, or description changed	Diff intervention text for existing arms	Sci, BD	Possible molecule pivot. May trigger reclassification.
<code>endpoint_changed</code>	Primary outcome measure modified	Diff <code>primaryOutcomes</code> list: text similarity < threshold	Sci	May change which analytes/biomarkers need measurement.
<code>sponsor_changed</code>	Lead sponsor or responsible party changed	Diff <code>leadSponsor.name</code> or <code>responsibleParty</code>	BD, Mkt	New business contact. May change addressable market classification.
<code>sites_changed</code>	Study locations added or removed	Diff locations list	Ops	Geographic expansion = more sample shipping logistics.

TIER 3 — SYSTEM

Category	What Changed	Detection Method	Teams	Business Signal
<code>metadata_only</code>	Contact info, IRB, dates, cosmetic text	No Tier 1/2 field changes — only non-business fields differ	—	No business impact. Filtered out of reports. Kept for audit.
<code>reclassified</code>	ALEXIS TA or modality changed (policy update)	Master TA/modality ≠ pulse, but no CT.gov field diff	Sci	Internal improvement. Not a real-world trial change.

2. Direction Tags

Every category with a direction captures which way the change went. '5 trials increased enrollment this week, 2 decreased' is more useful than '7 enrollment changes.'

Category	↑ Positive / Forward	↓ Negative / Backward
status change	not_yet → recruiting, suspended → recruiting	recruiting → suspended, any → terminated/withdrawn
completed	completed_passed	completed_failed
enrollment	enrollment_up (+N, +%)	enrollment_down (-N, -%)
intervention arms	arm_added (new molecule)	arm_removed
endpoints	endpoint_added	endpoint_removed
sites	sites_added (+N)	sites_removed (-N)
sponsor	sponsor_changed, collaborator_added	collaborator_removed
intervention text	drug_name_changed, dose_changed	— (all neutral, flag for review)
phase (linked)	phase_1_to_2, phase_2_to_3, phase_3_to_4	— (phase doesn't go backward)

Output schema per change:

```
{
  "category": "enrollment_change",
  "tier": 2,
  "direction": "enrollment_up",
  "old_value": 120,
  "new_value": 280,
  "delta": +160,
  "delta_pct": "+133%",
  "field": "enrollmentInfo.count"
}
```

3. Phase Linkage Engine

■ **THE PROBLEM:** When a drug advances from Phase 1 to Phase 2, the sponsor registers a **new NCT ID**. The Phase 1 trial shows status → 'Completed' and a separate Phase 2 trial appears as a new registration. Without linkage, these look like unrelated events.

Linkage fields (match across NCT IDs):

Field	Match Type	Weight	Notes
Sponsor name	Fuzzy (Pfizer Inc → Pfizer)	Required	Normalize common variations (Inc, Ltd, Corp)
Drug / intervention name	Fuzzy + alias resolution	Required	Same molecule: INN, brand, and code names (e.g., pembrolizumab = Keytruda = MK-3475)
Condition / disease	MeSH tree overlap	Strong	'breast cancer' and 'HER2+ breast cancer' should link
Phase	Ordered (new must be ≥ old)	Required	Phase 2 can only advance from Phase 1 or Phase 1/2
Completion → registration timing	Window (0–18 months)	Supporting	Phase 2 typically filed 3–12 months after Phase 1 completes
Modality	Exact	Supporting	Mismatch is a flag, not a disqualifier (could be reformulation)

Confidence scoring:

High (≥ 0.85)

Match: Sponsor + drug + condition + phase order + timing all match

Action: Auto-link. Tag as phase_advance.

Medium (0.60–0.84)

Match: Sponsor + drug match but condition broader/narrower, OR timing outside window

Action: Flag for review. 'Probable phase advance' in changelog.

Low (< 0.60)

Match: Partial — sponsor matches but drug doesn't (could be alias), or drug matches but sponsor differs (licensing)

Action: Do not auto-link. Log for manual review.

Drug name alias challenge: Same molecule can have multiple names (pembrolizumab = Keytruda = MK-3475). Bootstrap an alias table from MeSH supplementary concept records + UNII + ChEMBL. Start with top 500 drugs by trial count — covers ~80% of industry trials.

4. Outcome Assessment (Completed Trials)

When a trial completes, the key question is: did it work? A passed trial means a phase advance is coming (new contract). A failed trial means the program may be dead (no follow-on).

PASSED

Signal	Confidence	Source	Timing
New trial with same drug at later phase appeared	High	Linkage engine	0–6 months
resultsSection: 'met primary endpoint'	High	CT.gov resultsSection	6–12 months (lag)
Sponsor press release announcing advancement	High	Web search (future)	Days to weeks
Enrollment target met (actual $\geq$ target)	Medium	Diff enrollmentInfo	Immediate

FAILED

Signal	Confidence	Source	Timing
No new trial for same drug after 12+ months	Medium	Linkage engine (absence)	12+ months
resultsSection: 'did not meet primary endpoint'	High	CT.gov resultsSection	6–12 months
whyStopped: efficacy/futility language	High	CT.gov whyStopped	Immediate
Enrollment significantly below target	Low	Diff enrollmentInfo	Immediate
Same drug appears in new Phase 1 (reformulation)	Medium	Linkage engine	3–12 months

UNKNOWN

Signal	Confidence	Source	Timing
Too early — completed < 6 months ago	—	Time-based	Wait and reassess
Non-efficacy study (PK, biomarker, etc.)	—	study_intent field	N/A for pass/fail

Assess in waves:

Wave 1 (immediate): Check whyStopped + enrollment actual vs. target.

Wave 2 (monthly): Run linkage engine — did a later-phase trial appear?

Wave 3 (quarterly): Check resultsSection for trials completed 6–12 months ago.

Ongoing: Trials in 'unknown' reassessed every wave. 18+ months with no signal → presumed failed.

5. Processing Pipeline

Runs after each pulse, before the weekly dashboard generates. 8 steps, fully automated except for medium-confidence linkage review.

Step 1: Load & Index

Load master DB and pulse snapshot. Build NCT ID index on master for O(1) lookup.

IN: master_DB.json + pulse .json

OUT: master_index: dict[nct_id → trial], pulse_trials: list[trial]

Step 2: Split New vs. Existing

Check each pulse NCT ID against master index. Split into two lists.

IN: pulse_trials + master_index

OUT: new_trials (tag: new_registration), existing_trials

Step 3: Field-Level Diff

For existing trials, diff each monitored field. Record old value, new value, field name.

IN: existing_trials + master_index

OUT: diffs: list[{nct_id, field, old_val, new_val}]

Step 4: Categorize Changes

Map diffs to taxonomy categories. Apply direction tags. Multiple categories per trial.

IN: diffs per trial

OUT: trial.update_categories: list[{category, direction, tier, fields}]

Step 5: Linkage Engine

For new registrations + completed trials, match sponsor + drug + condition across NCT IDs.

IN: new_trials + recently_completed

OUT: linked_pairs: list[{old_nct, new_nct, phase, confidence}]

Step 6: Outcome Assessment

For completed trials, evaluate pass/fail signals across all available sources.

IN: completed_trials + linkage + CT.gov metadata

OUT: trial.outcome: {assessment, signals, confidence}

Step 7: Tag & Save

Attach all enrichment fields to pulse trials. Save enriched snapshot.

IN: All pipeline outputs

OUT: enriched_pulse_snapshot.json

Step 8: Generate Changelog

Aggregate by tier and category. Produce structured report.

IN: enriched_pulse_snapshot

OUT: changelog_report (.json + printable summary)

6. Output Format

Enriched trial schema (new fields added to each pulse trial):

```
{
  "nct_id": "NCT09182001",
  "therapeutic_area": "Oncology",
  "drug_modality": "gene_therapy",
  // ... existing ALEXIS fields ...
  "update_type": "existing",      // "new" | "existing"
  "update_categories": [
    {
      "category": "status_to_recruiting",
      "tier": 1,
      "direction": "from_not_yet",
      "old_value": "NOT_YET_RECRUITING",
      "new_value": "RECRUITING",
      "field": "overallStatus"
    }
  ],
  "linked_from": {
    "nct_id": "NCT08871234",
    "phase": "PHASE1",
    "link_confidence": 0.91
  },
  "outcome": {
    "assessment": "passed",
    "confidence": "high",
    "signals": ["Phase 2 registered by same sponsor"]
  }
}
```

Changelog report structure:

Section	Contents
Header	Window dates, total pulse trials, drug trials, new vs. update split
Tier 1 Summary	New registrations (N industry), status changes (recruiting, completed w/ outcome, terminated), new arms, phase advances
Tier 2 Summary	Enrollment changes (N up, N down), intervention modifications, endpoint changes, sponsor changes, site changes
Tier 3 Summary	Metadata-only count, reclassification count
Detail Tables	Full trial-level rows: NCT ID, sponsor, TA, modality, category, direction, old/new values — grouped by tier then category

Fields monitored for diff:

Category	CT.gov Fields	Maps to
Status	overallStatus, lastKnownStatus	Tier 1
Interventions	interventions[].name, .description, .type	Tier 1/2
Enrollment	enrollmentInfo.count, .type	Tier 2
Endpoints	primaryOutcomes[].measure, .timeFrame	Tier 2
Sponsor	leadSponsor.name, responsibleParty, collaborators[]	Tier 2
Sites	locations[].facility, .status	Tier 2
Dates	startDate, completionDate, primaryCompletionDate	Tier 3
Admin	contactsLocations, ipdSharing, oversightModule	Tier 3

Feeds into: Weekly Pulse Dashboard (BD action table, Ops velocity tracker, Science signal feed)

Dependencies: Master DB (diffing), MeSH supplementary concepts (drug aliases), CT.gov API v2 (resultsSection, whyStopped)

Build order: Taxonomy + diff engine first → linkage engine second → outcome assessment third (ship in waves)